TRAN TRUNG DUC

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Professional Summary

Self-directed **Machine Learning Engineer** with a strong foundation in **Deep Learning architectures (CNNs, Transformers)** and **Generative AI**. Experienced in building end-to-end ML pipelines, from data processing to model evaluation. Proficient in **Python** and **Prompt Engineering** (Llama 3), with practical exposure to **Computer Vision** products (SNOW App). Eager to join **Intuition Machines** to apply research mindset and engineering rigor to build scalable security and AI solutions in a global remote environment.

Education

University of Science, VNU-HCM Ho Chi Minh City
Electronics and Telecommunication Engineering | **GPA: 3.4/4.0 (Top Tier)** 2021–2025
*Relevant Coursework: **Linear Algebra**, **Probability & Statistics**, Matrix Calculus, Digital Signal Processing (DSP), Neural Networks, Python Programming.*

Technical Skills

- **Machine Learning & CV: Deep Learning** (CNN, LSTM, Attention Mechanisms), **Model Evaluation** (Loss functions, Bias-Variance tradeoff), TensorFlow/Keras, PyTorch basics.
- **Generative AI: Prompt Engineering** (LLMs), Fine-tuning concepts, API Integration (OpenAI, HuggingFace), Generative Algorithms awareness.
- **Software Engineering: Python** (Advanced - Clean Code, Unit Testing), Git/GitHub (CI/CD), Docker, Linux Environment.
- **Soft Skills: Remote Work Ready** (Self-directed), Technical Communication (English), Rapid Iteration Mindset.

Experience

Software Engineering Intern (ML Pipeline Focus) — ITR Vietnam Nov 2024 – Jan 2025
- **ML Pipeline Implementation:** Built automated pipelines to process and normalize large-scale datasets for AI models. Implemented **Unit Tests** for data processing functions to ensure pipeline reliability and data quality.
- **Code Optimization:** Optimized Python scripts for data transformation using Vectorization (NumPy), reducing processing latency by **40%**, demonstrating awareness of performance constraints in ML systems.
- **Collaboration:** Worked in an Agile environment, participating in code reviews and technical discussions to align ML implementation with system requirements.

QA / Product Collaborator (Computer Vision Product) — SNOW Vietnam Corporation Apr 2025 – Oct 2025
- **Visual Model Evaluation:** Conducted quality assessment for **AR Filters** and **Image Processing effects**. Evaluated model performance on edge cases (low light, occlusion) to ensure high-quality visual output for millions of users.
- **Feedback Loop:** Collaborated with the CV research team to report recurring artifacts in image generation, providing detailed technical feedback to refine the underlying algorithms.

Key Projects

Deep-PPG-Research-Core (Deep Learning Framework Development) *Github Project Link*
- **Deep Learning Architectures:** Designed and trained **CNN-LSTM** and **Transformer** models. Applied core Deep Learning principles (Convolutions, Attention Mechanisms, Backpropagation) that are directly transferable to **2D Computer Vision** tasks.
- **Rigorous Evaluation:** Implemented custom **Loss Functions** and evaluation metrics (MAE, RMSE) to benchmark model performance. Analyzed **Bias-Variance tradeoffs** to prevent overfitting on the MIMIC-III dataset.
- **Research Engineering:** Structured the project as a reproducible research framework with clear documentation, modular code design, and version control, ensuring maintainability.

Automated Processing Tool (GenAI & Prompt Engineering) *Github Project Link*
- **Generative AI Application:** Built a pipeline integrating **Whisper** (Audio-to-Text) and **Llama 3** (LLM). Applied advanced **Prompt Engineering** techniques to ensure accurate context understanding and translation quality.
- **Integration & Deployment:** Containerized the application using **Docker** for consistent deployment. Wrote maintainable Python code with error handling for API rate limits and network failures.

Cuffless-BP-IoT-System (End-to-End Data Pipeline) *Github Project Link*
- **Data Acquisition Strategy:** Designed the full data lifecycle from sensor acquisition to model inference. This demonstrates the ability to handle the "dirty work" of ML: data collection, cleaning, and preprocessing.
- **Edge ML Deployment:** Deployed optimized lightweight models (TFLite) for inference on mobile devices, adhering to strict latency and resource constraints.

Certifications and Awards

- **VPBank Technology Hackathon 2025** (System Design)
- **Biomedical Engineering Innovation Competition 2025** (Certificate)
- **Coursera:** ML with Python • Deep Learning (Keras) • DL (TensorFlow)